

MINGLE: Learning Whole-Body Humanoid Social Navigation from Paired Egocentric Perception and Motion-Captured Actions

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Abstract—Social navigation requires more than collision avoidance: A robot must move in ways people can anticipate and respond to. Existing approaches largely reduce navigation to a 2D path executed through velocity commands. For humanoids, this abstraction overlooks articulated geometry, legible whole-body motion, and biped locomotion dynamics.

We propose MINGLE, which learns whole-body humanoid social navigation from paired human demonstrations: Egocentric RGB-D records what the navigator perceives, while motion capture records the whole-body action taken in response.

Using a portable capture rig, we collect 272 episodes across five interaction families and retarget the motions to a Unitree G1. A geometry- and pedestrian-conditioned diffusion policy combines a history bird’s-eye-view of scene geometry with the pedestrian’s articulated motion to predict short-horizon whole-body actions and longer-horizon navigation waypoints, executed by a general-purpose whole-body controller.

Experiments in simulation and on a real Unitree G1 show that MINGLE improves safety and navigation efficiency over conventional velocity-based social navigation, enabling the humanoid to negotiate shared space through whole-body motion. More videos are available at mingle-nav2027.github.io.

I. INTRODUCTION

A crowded corridor is a conversation in motion (Fig.1). People do not merely avoid one another: They reveal intent, commit to a side, adjust speed and spacing, and expect others to do likewise [1]. Humanoids entering corridors, offices, and homes must learn to take part in this exchange. A robot that simply moves slowly enough for everyone else to walk around it may remain collision-free, but it is an obstacle rather than a participant. Social navigation requires both parties to adjust so that each can continue naturally.

Existing robot social navigation has been developed for wheeled and quadruped platforms. The prevailing formulation represents navigation as a path or direction in 2D space, then executes it through velocity commands [2]–[5]. This abstraction captures where the robot should move.

For a humanoid, *where to move* is only part of the decision. Three factors make this abstraction incomplete: *articulated geometry*, *legible intent*, and *locomotion dynamics*.

First, *articulated geometry matters because a humanoid is not a rigid footprint*. Consider two people passing in a narrow corridor. Choosing opposite sides determines their paths, but not the maneuver: They may turn their shoulders, rotate their torsos, tuck their arms, shorten a step, or step sideways to create room. A 2D path specifies where to go, but not how an articulated body should fit through the available 3D space.

Second, *whole-body motion makes intent legible*. The pelvis may continue forward while the torso turns toward the chosen passing side; the shoulders may rotate before a



Fig. 1. MINGLE navigates narrow scenes with whole-body social behavior learned from human demonstrations. A real-world encounter in a 1.3 m-wide passage, with the robot’s forward speed labeled on each frame. The robot first commits to one side to show its intention, then slows to a stop while rotating body and closing arms to make room, and resumes walking once the pedestrian passed. Unlike general velocity tracker that overshoots in sharp turns at similar speed, MINGLE achieves safe, collision-free performance against both static obstacles and pedestrians.

lateral displacement becomes pronounced. Such posture and motion provide cues about what comes next. A direction of travel alone does not specify this postural language.

Third, *locomotion dynamics constrain how a biped can react*. A humanoid cannot turn, accelerate, or decelerate arbitrarily while maintaining balance. EmoBipedNav [6] commands sagittal speed and heading, with speed capped at 0.4 m/s, well below the 1.3 m/s to 1.5 m/s comfortable walking speeds of healthy adults [7]. At such speeds, pedestrians may simply walk around the robot, reducing social navigation to obstacle avoidance rather than mutual negotiation between peers.

The problem is thus not only *where the humanoid should move*, but also *how it should compose its body while doing so*. Socially appropriate answers are difficult to prescribe. Model-based methods encode pedestrian dynamics and interaction rules [2], [8], while reinforcement-learning methods optimize designed objectives [3]–[5]. Both can produce effective dynamic collision avoidance, but social behavior is not just about clearance. How people pass – *which side they choose, when they yield, how decisively they commit, and how much space they leave* – often follows local conventions.

Human development offers a useful cue. Imitative learning is a foundational form of human cultural learning [9], and imitation also serves a social function through which children learn the conventions and rituals of their groups [10]. In experiments with children aged three to six, behaviors presented as conventional are imitated and transmitted with greater fidelity than the same behaviors presented as merely

instrumental [11]. This suggests a different route for humanoid social navigation: *Learn how people negotiate shared space by observing and imitating how they actually do it.*

Learning this way raises a data question: How can a humanoid acquire enough social experience? Collecting close interactions directly with a humanoid is difficult to scale, particularly before it has learned socially competent behavior. Humans, by contrast, negotiate such encounters routinely. Recording what they perceive together with how their whole bodies respond can turn these everyday interactions into demonstrations for a humanoid.

Existing datasets provide only pieces of this experience. Social-navigation datasets pair observations with planar trajectories [12], [13] or head motion [14], discarding much of the articulated motion available to a humanoid. Human-motion archives provide rich whole-body motion but not the mover’s ego-centric perception [15], while large ego-centric datasets do not provide the paired whole-body actions required for social navigation [16], [17]. To our knowledge, no existing dataset pairs ego-centric perception with full-body motion during social navigation.

We thus propose **MINGLE**, short for *Motion Imitation for NaviGation Learned from Egocentric Perception* (Fig. 1). The framework proceeds in three steps. **1) Capture human social navigation experience.** A portable inertial MoCap suit and chest-mounted stereo camera record ego-centric RGB-D perception and whole-body motion simultaneously as a person negotiates corridors, doorways, intersections, and overtaking encounters. **2) Turn human experience into humanoid demonstrations.** We retarget the demonstrator’s whole-body motion to a Unitree G1 and recover the interacting pedestrian’s articulated pose from the same ego-centric video. Each episode therefore pairs the observed scene with a robot-compatible whole-body action and the motion of the person to whom that action responds. **3) Learn whole-body social reactions.** A conditional diffusion policy combines the humanoid’s recent state, a history bird’s-eye-view (BEV) representation of the surrounding geometry, and the pedestrian’s articulated motion to predict short-horizon whole-body motion together with longer-horizon navigation waypoints. A general-purpose whole-body tracking controller [18] executes the predicted motion on the humanoid.

Our work makes three major contributions. **1) A scalable dataset for whole-body social navigation:** We collect 272 episodes across five interaction families, pairing ego-centric RGB-D observations with full-body human motion retargeted to a Unitree G1 using a portable capture rig that can be deployed scalably in new environments. **2) A geometry- and pedestrian-conditioned whole-body navigation policy:** The policy combines a history BEV representation of surrounding geometry with the interacting pedestrian’s articulated motion, and jointly predicts short-horizon whole-body motion and longer-horizon navigation waypoints. **3) Whole-body social navigation on a real humanoid:** We demonstrate zero-shot deployment on a Unitree G1 and evaluate the proposed whole-body interface against conventional velocity-based social navigation in simulation and on hardware.

II. RELATED WORK

Social Navigation and Learning from Human Demonstration. To learn socially compliant navigation behavior, studies have been divided into two major areas. The first is to model the reactive pedestrian responses [19]–[21]. Then optimization-based planners are used to find optimal solution for the collision-free behaviors, either through model-based approaches [2], [8], [22], or through reinforcement learning like [3]–[6]. However, it is hard to model human reactions in social environment, and these methods still take social navigation as dynamic collision avoidance problem where the social compliance is missing. The second direction comes from data-driven methods. Researchers utilize teleoperated or human demonstration to learn social navigation skills. Datasets [12]–[14] record socially compliant motion and reduce the demonstration to a trajectory to follow so that many works [23]–[26] utilized these datasets for wheeled and quadruped robots, but not humanoid. Conversely other datasets [15]–[17] hold whole-body human motion in abundance but carry no navigation tasks. To the best of our knowledge, our dataset offers a novel pairing of ego-centric observation with whole-body action for social navigation.

Humanoid Whole-Body Control. Even if full-body motion is present, a low-level controller that takes in the motion plan and returns an executable command for robots is inevitably needed. PHC [27] made this practical for simulated avatars, and methods [28]–[31] carried it to hardware as a real-time interface over retargeted human motion. SONIC [18] scales this to a hundred million frames of training motion, yielding one generalist policy that can track motion references in real time. We build directly on SONIC to execute our planned motion on the G1 robot.

Diffusion Policies. Diffusion models [32], [33] suit robot policies because of its multi-modality. Diffusion Policy [34] established this for manipulation, and MDM [35] and CLoSD [36] for whole-body human motion. NoMaD [37] is the closest navigation instance, over a planar action space. Diffuse-CLoC [38] utilizes diffusion policy with guidance to output whole-body robot actions to tackle collision avoidance and other tasks. [39] also utilizes a diffusion policy to output whole-body motion for vision-language based navigation. Ours differs in acting on whole-body motion while not only focusing on static collision avoidance, but also how to maintain social awareness with the human in the scene.

III. PROBLEM FORMULATION

Our goal is to use imitation learning to let a humanoid navigate challenging environments around people while avoiding collisions and following social norms. We formulate it as a goal-free policy learning problem. The policy receives RGBD perception O_t and a length- T proprioceptive history $S_t = s_{t-T}, \dots, s_t$, and outputs an H -step action chunk $a_{t, \dots, t+H}$, where a_t specifies the full-body joint positions and root rotation for controlling a Unitree G1.

Demonstration

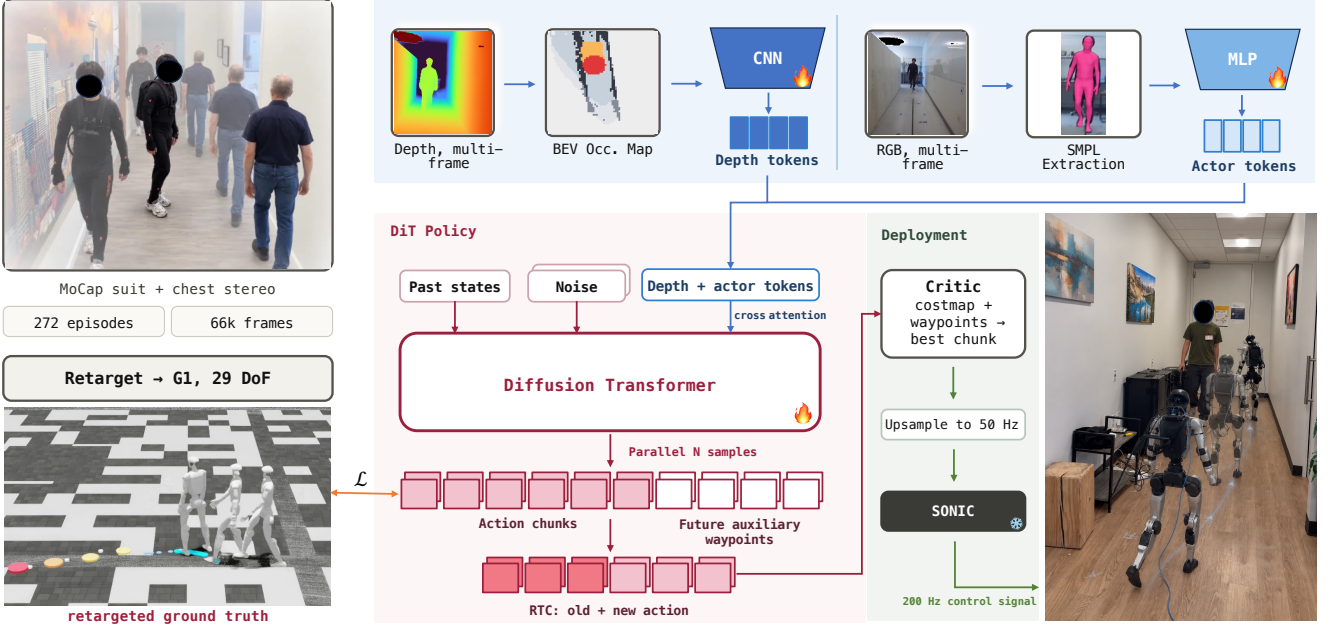


Fig. 2. **Overview of MINGLE.** Human demonstrations captured with a MoCap suit and a chest-mounted stereo camera are retargeted onto the G1. At run time the policy consumes a history BEV occupancy map and a window of estimated pedestrian pose, denoises several 30-frame whole-body chunks in parallel. After RTC [40] and critic rating, the best chunk will be executed using SONIC [18], achieving closed-loop deployment in different scenes.

IV. METHOD

MINGLE learns a whole-body social navigation policy directly from human demonstrations of the task. A person wearing a motion-capture suit and a chest-mounted stereo camera walks through different social scenarios encountering other pedestrians. Retargeting the suit onto a Unitree G1 turns each recording into a paired (observation, whole-body action) demonstration for the robot (Sec. IV-A). A conditional diffusion policy is then trained to denoise a one-second chunk of 29-DoF joint targets together with the planar root motion that carries them, conditioned on two perception branches and on the robot’s own recent state (Sec. IV-B). Fig. 2 gives the overall picture.

A. Data Collection and Processing

Capture rig. Demonstrations are recorded by a single operator wearing an Xsens motion-capture suit and carrying a ZED 2i stereo camera at chest height. The suit streams full-body kinematics and the camera records RGB-D data. We deliberately kept the recording minimal. There are no external cameras, no markers on the walls and no map. Therefore, the setup can be carried into arbitrary scenarios.

Retargeting. Each MoCap take is retargeted onto the 29-DoF Unitree G1 with a motion retargeter [41], which solves for the joint configuration that best reproduce the human’s link frames subject to the robot’s joint limits. The result is a robot trajectory $\{q_t \in \mathbb{R}^{29}, T_t \in \text{SE}(3)\}$ at 30 Hz that is kinematically feasible for the G1 by construction.

Pedestrian state. The human actor is recovered from the same ego-centric video. We run a monocular 4D human reconstruction [42] on the RGB stream to obtain per-frame

TABLE I
DATASET COMPOSITION.

Family	Episodes	Interaction
Corridor	92	head-on encounter
Doorway	48	0.7 m gap, one must yield
Intersection	42	occluded crossing
Overtake (slow)	52	same-direction pass
Overtake (obst.)	38	pass with narrowed lane
Total	272	66,026 frames, 36.7 min

SMPL [43] parameters and 3D joints, lift them into the camera frame. Similar to prior human motion generation works [36], [44], [45], each frame yields a 216-dimensional actor vector: the global orientation and 23 body-pose rotations in 6D rotation parameterization and 24 camera-frame joint positions. No-human-detected frames are flagged, allowing explicit handle (Sec. IV-B.3).

Dataset. The released set contains 272 episodes (66,026 frames, 36.7 min at 30 Hz) spanning five interaction families: *corridor* head-on encounters (92), *doorway* negotiations through a gap that fits one body (48), *intersection* crossings where the pedestrian is occluded by a corner until late (42), and two *overtaking* families in which the pedestrian moves the same way as the operator, either slowly (52) or while an obstacle narrows the passing lane (38). We hold out $\approx 12\%$ of *each* family at random, giving 241 training and 31 validation episodes, so no family is validated more thinly than another (Table I). All normalization statistics are recomputed over the 241 training episodes alone.

B. Policy Architecture

1) *Action representation:* At every replanning instant the policy emits a chunk of $H = 30$ future frames (1.0 s at 30 Hz).

TABLE II
CHANNELS OF THE HISTORY BEV MAP $\mathbf{M}$.

	<i>current</i>		<i>remembered</i>
0	static occupancy	4	static, pooled
1	max height	5	free, pooled
2	observed free	6	pedestrian, pooled
3	pedestrian		

A single frame of the chunk is

$$x_k = \left[\underbrace{q_k}_{29} \mid \underbrace{\psi_k}_6 \mid \underbrace{p_k}_2 \right] \in \mathbb{R}^{37}, \quad k = 1 \dots H, \quad (1)$$

where q_k are joint positions, ψ_k is the root rotation expressed in rot6d [46] representation and p_k the planar position.

Both ψ_k and p_k are expressed with respect to the *last observed frame*. For ψ_k , we will calculate the yaw segment relative to anchor, while keeping both roll and pitch in global frame. This representation is directly reusable at deployment where only relative yaw is observable. Note that we never use the predicted xy to *control* the robots. We only use the full joint actions with the root rotation, while the predicted xy is used as an auxiliary output for critic only.

2) *Perception branch I: History Bird-Eye-View*: Note that we have a relatively small dataset. Directly training a depth encoder from scratch often results in overfitting. Therefore, an intermediate representation derived from depth that can generalize across different scenarios is necessary. We propose to use an ego-centric, gravity-aligned *history* BEV occupancy map $\mathbf{M} \in \mathbb{R}^{7 \times 64 \times 64}$ covering $[-2, 5]$ m forward and $[-3, 3]$ m laterally, with the seven channels listed in Table II. Channels 0–2 are rasterized from the gravity-aligned point cloud of the current depth frame. A cell is marked occupied if its maximum point height falls in the $[0.22, 2.0]$ m band. Channel 3 holds the pedestrian’s footprint and is *carved out* of channel 0. Channels 4–6 are the memory: $K=8$ past frames at stride 10 are rigidly warped into the current frame by their measured SE(2) ego-motion and max-pooled with a 0.9^k decay. The 7-layer BEV map is then passed through a small strided CNN, adaptively pooled to an 8×8 lattice and projected to the model width, generating 64 map tokens.

3) *Perception branch II: actor motion*: The second branch reads the pedestrian’s articulated motion directly rather than through the occupancy grid. The last 10 actor vectors, sampled at stride 3, are normalized and passed through an MLP to 10 memory tokens, with learned positional embeddings.

Two mechanisms handle when the pedestrian is *not* observed. A learned *missing token* replaces the MLP output on frames where the tracker detected no one. Substituting a free parameter after the MLP lets “no observation” be learned as its own embedding. Second, with probability $p_{\text{drop}}=0.15$ the entire actor window is replaced by the missing token during training even when a person *was* detected. This trains the unconditional mode needed for classifier-free guidance on the actor stream at inference.

4) *Diffusion policy*: Let $\mathbf{X}_0 = [x_1 \dots x_H \mid w_1 \dots w_K]$ be the clean action chunk augmented with $K=4$ long-horizon waypoints (below). A cosine-schedule DDPM with $T=100$

steps corrupts it, and the network is trained to denoise the corrupted sample. We adopt Diffusion Transformer as our policy architecture [47]. The denoiser is a transformer *decoder* whose target is the concatenation $\text{tgt} = [\mathbf{P} \mid x_{1:H} \mid w_{1:K}]$ of the clean 4-frame past proprioceptive states with the noisy chunk and waypoint, cross-attended with perception conditions. A block-directional mask stops the past prefix from attending to the future, so the one part of the target that is known exactly is never corrupted by mostly noise; the future block attends to everything.

Long-horizon waypoints. One second of lookahead is enough to execute a maneuver but not to decide on one. Whether to pass a pedestrian on the left or the right is a three-second question. We therefore append $K=4$ *root-only* waypoint tokens at $+45, 60, 75, 90$ frames (1.5–3.0 s), carrying only $[p \mid \psi] \in \mathbb{R}^8$ in the same anchor-relative convention as a chunk row, with their own input projection and output head. The attention between short-horizon motion and the long-horizon waypoints is fully bidirectional. Their positional encoding is placed at *true time* instead of index.

Normalization. we follow the statistics of each quantity to normalize. Joint positions and velocities are mapped to $[-1, 1]$ through the robot’s limits. Planar displacement is min–max normalized *per horizon step*, because its spread grows by more than an order of magnitude between $k=1$ and $k=H$. Heading is also zero-mean normalized per-step. **Training Objective.** With $\hat{\cdot}$ denoting the prediction and all terms per-dimension means,

$$\mathcal{L} = \underbrace{\|\hat{q} - q\|^2}_{\text{joints}} + \lambda_\psi \|\hat{\psi} - \psi\|^2 + \lambda_{xy} \|\hat{p} - p\|^2 + \lambda_{wp} (\lambda_{xy} \|\hat{w}^p - w^p\|^2 + \lambda_\psi \|\hat{w}^\psi - w^\psi\|^2), \quad (2)$$

with $\lambda_\psi = 3$, $\lambda_{xy} = 1$ and $\lambda_{wp} = 0.5$. Remaining training details are reported with the experiments in Sec. V-A.

5) *Deployment*: At run time the policy is sampled with DDIM and re-planned on a fixed quantum. We sample multiple trajectories at a time and choose the best trajectory with respect to the BEV occupancy map. We rate them through a distance-transform costmap similar to [48]. Write $\mathbf{x} \in \mathbb{R}^2$ for a point in that frame, $d_s(\mathbf{x})$ for the Euclidean distance transform of the static occupancy (channels 0 and 4 thresholded at $\tau=0.2$), d_h for the same on the pedestrian channel. We estimate the future possible pedestrian space using constant-velocity forecast, resulting in d_f . Let Ω be the cells the robot has evidence about. The cost of a point is

$$C(\mathbf{x}) = \max \left(e^{-\bar{d}_s/\lambda_d}, e^{-\bar{d}_h/\lambda_h}, w_f e^{-\bar{d}_f/\lambda_h}, \mathbf{1}[\mathbf{x} \notin \Omega] \right) \in [0, 1]. \quad (3)$$

with two length scales because a wall and a person are given way at different radii (λ_d and λ_h are 0.5 m and 0.8 m). The forecast term is discounted ($w_f=0.7$). The last term charges unobserved space: a chunk that steers into the part of the map the robot has no clue should lose to one that does not.

The bars denote body inflation, since the G1 is 0.4 m wide and a point rating would score a center line 0.15 m from a wall as merely expensive. As a distance transform is 1-Lipschitz and $e^{-d/\lambda}$ monotone, the worst case over a disc of

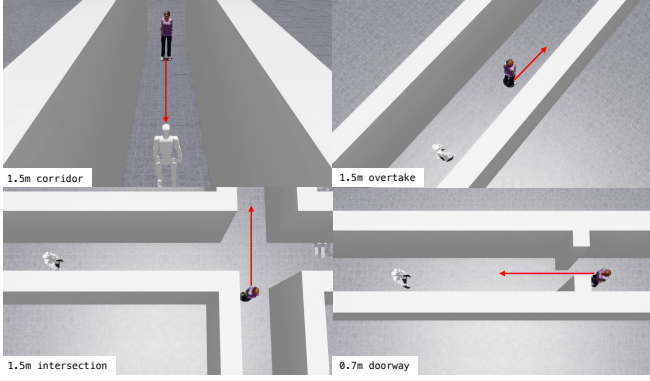


Fig. 3. **Four simulated evaluation scenarios** Each panel is a top-down view of a scene built from primitive geometry and labeled with its passage width. The red arrow shows the pedestrian’s scripted walking direction. The scenarios span a head-on corridor encounter, overtaking a slow pedestrian, a crossing at an occluded intersection, and a 0.7 m doorway that only one of the two can pass at a time. Each represents a different social decision.

radius r is exactly $e^{-\bar{d}/\lambda}$ with $\bar{d} = \max(d - r, 0)$, so C reaches 1 precisely when the body touches an obstacle.

The selected chunk is handed to a whole-body tracking controller, SONIC [18] in our case, which converts the 29 joint targets plus root pose into torques; because the chunk is a *motion* and not a twist, the controller is free to use the arms and torso to stay balanced through a maneuver a velocity command could not express. Sampler settings and rates are given with the implementation details in Sec. V-A.

V. EXPERIMENTS

We evaluate our method on Unitree G1 robots in both simulation and real world to answer three questions: **(Q1)** Does predicting whole-body motion outperforms the twist commands that social navigation planners conventionally produce (Sec. V-C)? **(Q2)** Which components of our pipeline account for the performance (Sec. V-D)? **(Q3)** Does the result transfer to real world (Sec. V-E)?

A. Implementation Details

MINGLE is deliberately small. The BEV branch uses a three-layer strided CNN ($64 \rightarrow 128 \rightarrow 256$) to turn the single $7 \times 64 \times 64$ image into an 8×8 grid of 64 tokens; the actor branch is a two-layer MLP over 10 pose tokens, dropped as a block with probability 0.15 during training. We utilize RF-DETR [49] for human detection and 4D-Humans [42] for SMPL estimation. Both sets of tokens feed to the cross attention layer of the 4-layer transformer decoder (width 256, feed-forward 512, 4 heads, dropout 0.1). We train the cosine-schedule diffusion ($T=100$, $\lambda_\psi=3$, $\lambda_{xy}=1$, $\lambda_{wp}=0.5$) for 75000 steps at batch size 32, with AdamW at 10^{-4} annealed by a cosine schedule to 10^{-6} , gradient clipping at 1.0, conditioning noise $\sigma=0.02$ as augmentation, and an EMA of the weights with decay 0.9999. We train and evaluate on a single RTX 5090. At deployment we sample with DDIM at 10 steps, draw $S=64$ trajectories in parallel, score them against the current BEV, and replan every 0.5 s. We upsample the joint actions to 50 Hz to feed to SONIC.

B. Metrics

We report 1) **Success (%)**: the trial ends with no human-collision event and no fall. 2) **Human collisions**: mean number of contiguous contact *events* with the pedestrian per trial. 3) **Scene collisions**: the same, for contact with walls. 4) **Peak contact force (N)**: the per-trial maximum contact force over wall contact, taking average over trials. 5) **Minimum clearance (m)**: closest robot–pedestrian center-to-center distance over the trial, a proxy for social-navigation comfort rather than a contact test. 6) **Distance (m)**: net progress along the corridor from the start. 7) **Falls (%)**: percentage of trials in which the robot falls.

C. Comparison against Social Navigation Baselines

Baselines. We compare against two open-source social navigation methods spanning the optimization/learning divide, both of which output a unicycle twist $[v, \omega]$: 1) **SICNav** [2], SICNav’s model-predictive controller with a constant-velocity pedestrian model at its default speeds (maximum 0.95 ms^{-1}). Ground-truth metric state of the pedestrian and the corridor walls as line segments are given. It plans a 1.0 s SE(2) horizon. 2) **EmoBipedNav** [6], a learned social navigation policy for bipeds (SAC-AE over stacked LiDAR occupancy images), capped at 0.4 ms^{-1} . Its observation is raycast from the same segment-and-circle scene. Neither baseline produces whole-body motion. We execute both with the same pre-trained velocity-tracking policy on the G1.

Simulation setup. All closed-loop evaluation runs on a Unitree G1 in four scenarios built from primitive geometry with two overtake combined (Fig. 3). *corridor* (1.5 m wide, pedestrian walking head-on), *doorway* (a partition with a 0.7 m gap that fits one body, so one of the two must yield), *intersection* (a side corridor whose mouth is hidden by the corner until the robot is 3 m out, at which point the pedestrian crosses), and *overtake* (a 2.0 m corridor with the pedestrian walking the same way at $\approx 0.3 \text{ ms}^{-1}$, so the robot must close and pass). The pedestrian is an animated character driven by a fixed command list and is not reactive. Each trial randomizes the robot’s spawn by up to $\pm 1.0 \text{ m}$ along and $\pm 0.3 \text{ m}$ across the corridor and $\pm 15^\circ$ in yaw, using the same seeds for every method so that trials are paired, and runs for 500 control steps (10 s) at 50 Hz. The episode will end immediately when a failure happens.

Results. Table III reports the comparison. Baselines diverge from MINGLE where space is tight, i.e. corridor and doorway. The velocity-control MINGLE isolates the velocity-control vs. whole-body-control interface. What it lacks is how the encounter is executed: mean peak contact force is much higher, and more scene collisions. The whole-body action is both buying more room, and shaping the body to fit the room it already has, which is hard for a twist to express. The two baselines’ designed speed differs with SICNav up to 0.95 ms^{-1} , and EmoBipedNav capped at 0.4 ms^{-1} , so distance traveled in a trial is not comparable.

Fig. 4 and Fig. 5 make this concrete on the corridor, the most telling of the four scenarios because the passage is barely wider than two bodies and the encounter cannot be

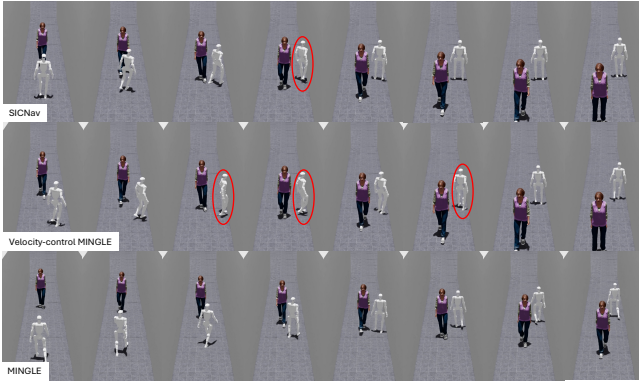


Fig. 4. **Qualitative comparison in the simulated corridor encounter.** Each row shows one method in the same head-on encounter. Red ellipses mark frames in which collision happens. Both velocity-controlled robots make contact while passing, whereas MINGLE turns its body to make room and passes without contact with the obstacle, showing that the gain comes from whole-body execution rather than from the planned path alone.

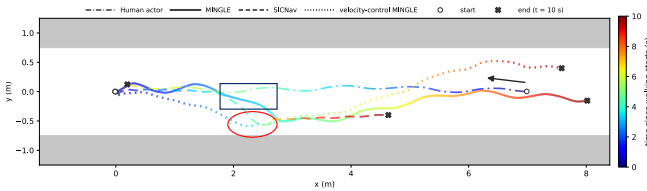


Fig. 5. **Top-down trajectories in the corridor encounter.** Line style identifies the agent and color encodes time since walking starts. Gray bands are the walls of the 1.5m corridor and the arrow gives the pedestrian’s walking direction. Blue box: SICNav walks till last moment to turn sharply to avoid collision with the pedestrian. Red oval: using general velocity tracker causes overshoot at turns around walls. Both methods actually collide with the walls due to the size of the body. MINGLE leaves enough social space early and achieves collision-free performance.

resolved by the path alone. We omit EmoBipedNav from the qualitative comparison for its poor results. In Fig. 4 both velocity-controlled methods contact the wall after turning, whereas MINGLE rotates its torso and passes cleanly in the same encounter. Fig. 5 shows how each methods gets there: SICNav holds the center of the corridor and turns sharply only at the last moment, and the velocity tracker overshoots as it turns back from the wall. Both then graze the wall causing collisions. MINGLE instead commits to one side well before the pedestrian arrives, keeps that offset through the pass, and covers the most distance.

D. Ablations

Setting. Every ablation method is generated from the deployed configuration by substituting only the fields that method changes, so the data lists, the train/validation split, the model size, the diffusion process, the optimizer and the schedule length are identical to the full model. All methods train for the same 75k steps on the same 241 episodes.

We vary four axes, one variable each. **1) Scene representation (DEPTH-CNN):** The history BEV occupancy map is replaced by the raw ego-centric depth images, encoded by a CNN trained from scratch on the same data. This asks whether projecting depth into an ego-centric occupancy map is necessary **2) Actor representation (ACTOR-2D):** The articulated actor stream is removed, so the pedestrian survives only as the coarse 2D footprint already rasterized

into the BEV occupancy channels. **3) The planar objective (NO-XY):** $\lambda_{xy}=0$ and the waypoint term removed together, so no planar quantity is supervised anywhere in the loss: the model keeps only the full-body motion and root rotation that SONIC consumes, and loses both the executed 1.0s path and the 1.5–3.0s auxiliary goals. **4) Planning horizon (HZ- $\{15,45\}$):** $H \in \{15,30,45\}$ frames (0.5–1.5 s), with the waypoint offsets shifted to stay outside each method’s chunk. $K=4$ and the 0.5 s spacing are held fixed.

Results. Table IV shows the results of these ablation methods. Replacing the occupancy map with a depth CNN is the most damaging change of all: DEPTH-CNN falls to 23.0% success with high pedestrian contacts. This overfits the appearance of the rooms the data was recorded in, while the BEV map presents similar geometry in different environment. The intermediate representation makes the policy generalize. Keeping only the pedestrian’s 2D footprint (ACTOR-2D) also degrades the performance. Where a person is bounds the collision problem; how they are moving is what says whether they are about to do. The hardware rollouts in Sec. V-E show an example of reading a gesture at the occluded intersection, which a footprint cannot express.

Supervising planar motion is what lets the policy tell good trajectories apart. Differing the horizon degrades the performance as well. HZ-15 will focus and learn mostly on how to act in short horizons, causing the drift in deployment. While HZ-45 has less fine details of execution.

E. Hardware Experiments

1) Platform. The policy is deployed on a Unitree G1 with a ZED2i camera mounted at chest height. Perception, the diffusion policy and the whole-body controller all run onboard: the BEV map is built from the live depth stream at 30 Hz, the pedestrian is tracked from the same stream, the policy replans every 0.5 s with DDIM, and the controller runs at 50 Hz. Depth is consumed in the same intrinsics the training data was recorded. **2) Policy.** Hardware trials use whole-body execution only. We do not repeat the velocity-control comparison on hardware, because the contact forces that method produces in simulation (Table III) are not ones we are willing to reproduce beside a person. **3) Scenarios and protocol.** We repeat the four interaction families on hardware with a human actor walking on his own will: head-on in a corridor 1.3m wide, a doorway 0.75m wide, a blind corner, and an overtake. Each configuration is run 10 times, alternating the starting position of both the robot and the pedestrian. **4) Results.** Table V reports the hardware quantitative results. MINGLE succeeds in 97.5% of the 40 trials. Fig. 6 shows the clips of the social behaviors learned. In a wide corridor the robot turns its torso and passes shoulder to shoulder instead of stopping. At the intersection it yields by default when someone appears, and when the person waves it through it reads that motion and resumes. This demonstrates the necessity of full actor motions. In both overtaking families it closes on the slower walker and passes on the free side, drawing its arms in when an obstacle narrows the lane.

TABLE III

CLOSED-LOOP COMPARISON AGAINST SOCIAL NAVIGATION BASELINES ON THE G1, AVERAGED OVER 50 PAIRED TRIALS PER SCENARIO. SICNAV IS SICNAV WITH A CONSTANT-VELOCITY PEDESTRIAN MODEL. THE BASELINES ARE EXECUTED THROUGH A VELOCITY-TRACKING CONTROLLER.

METRIC DEFINITIONS ARE GIVEN IN SEC. V-B.

Method	Low level	Success (%) $\uparrow$	Human coll. $\downarrow$	Fall $\downarrow$	Min. clear. (m) $\uparrow$	Peak force (N) $\downarrow$	Scene coll. $\downarrow$	Distance (m) $\uparrow$
		Mean (cor., door., int., over.)						
SICNav [2]	velocity	68.5 (68.0, 10.0, 96.0, 100.0)	0.27	0.05	0.32 (± 0.21)	937	1.06 (± 1.07)	5.85 (± 1.38)
EmoBipedNav [6]	velocity	66.0 (12.0, 52.0, 100.0, 100.0)	0.34	0	0.22 (± 0.07)	762	0.37 (± 0.66)	3.42 (± 0.71)
Vel-control MINGLE	velocity	94.5 (86.0, 92.0, 100.0, 100.0)	0.05	0.01	0.41 (± 0.13)	1145	1.78 (± 0.87)	6.29 (± 1.95)
MINGLE (ours)	whole-body	96.5 (92.0, 94.0, 100.0, 100.0)	0.04	0	0.40 (± 0.11)	174	0.40 (± 0.62)	7.19 (± 1.62)

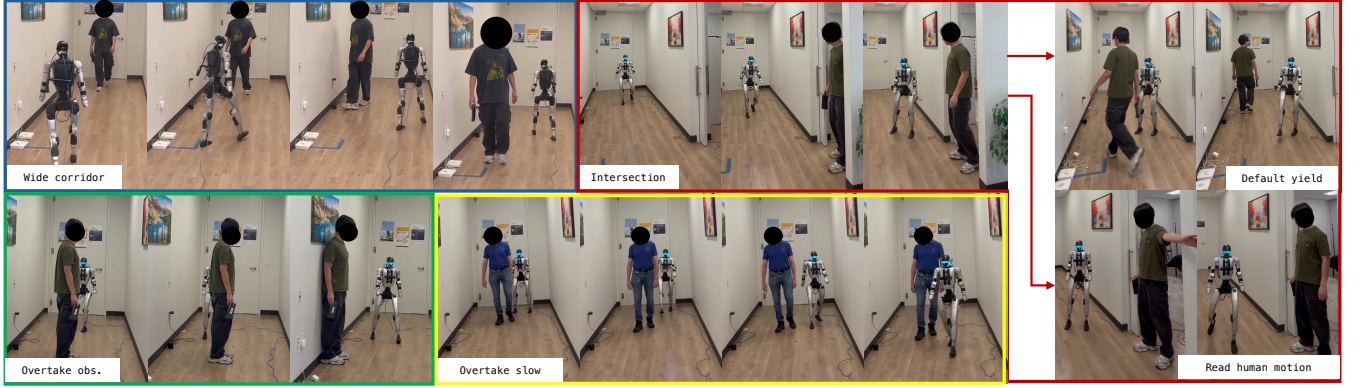


Fig. 6. **Whole-body social behaviors on hardware.** Each colored group is one encounter type on the Unitree G1, read left to right in time: passing in a wide corridor, an occluded intersection, overtaking around an obstacle, and overtaking a slower walker. The two panels at right expand the intersection into the two responses the policy produces there — yielding by default when the pedestrian appears, and reading the person’s motion as an invitation to proceed.

TABLE IV

ABLATION: CLOSED-LOOP CONSEQUENCE IN SIMULATION, AVERAGED OVER 50 TRIALS PER SCENARIO.

Method	Success (%) $\uparrow$	Human coll. $\downarrow$	Scene coll. $\downarrow$	Min. clear. (m) $\uparrow$
MINGLE	96.5	0.04	0.40 (± 0.62)	0.40 (± 0.11)
DEPTH-CNN	23.0	0.77	0.48 (± 0.77)	0.17 (± 0.35)
ACTOR-2D	61.0	0.39	0.37 (± 0.51)	0.24 (± 0.39)
NO-XY	47.5	0.53	0.86 (± 1.02)	0.21 (± 0.33)
HZ-15	35.5	0.65	0.15 (± 0.34)	0.19 (± 0.41)
HZ-45	87.5	0.13	0.35 (± 0.67)	0.37 (± 0.20)

TABLE V

HARDWARE RESULTS ON THE UNITREE G1. 10 TRIALS PER SCENARIO

Scenario	Success (%) $\uparrow$	Min. clear. (m) $\uparrow$	Contacts $\downarrow$
Corridor	100	0.31 (± 0.09)	0.2 (± 0.4)
Doorway	100	0.29 (± 0.07)	0.2 (± 0.4)
Intersection	90	0.47 (± 0.12)	0
Overtake	100	0.33 (± 0.15)	0
Mean	97.5	0.35 (± 0.21)	0.1 (± 0.3)

VI. CONCLUSION

We presented MINGLE, a pipeline that scalably collects paired ego-centric perception and full-body human demonstrations and trains a diffusion policy that produces whole-body humanoid motion for social navigation. Our experiments support two claims. **1)** Whole-body motion is a better control interface than velocity tracking for humanoid in narrow and challenging spaces for social navigation. **2)**

Imitation learning on our paired dataset generalizes through intermediate representations for new scenes. Due to limited time, one limitation is that the dataset size is small compared to other datasets. As future work, we would collect more social scenarios and interactions with several people.

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